

2.1 Categorical Variables

- Goal: We need to turn “raw” data into information.
- Data are summarized into
 - 1 tables & graphs
 - 2 numerical indices
- One categorical variable
 - 1 Summary statistics: frequency table, proportion
 - 2 Visualization: bar graph, pie chart
- Two categorical variables
 - 1 Summary statistics: two-way table, difference in proportions
 - 2 Visualization: segmented or side-by-side bar graph

One Categorical Variable

- For a categorical (qualitative) variable, frequency table is a basic technique for organizing the outcomes of the categorical variable.
- **Frequency table** lists all the **categories** of a categorical variable and **counts** how often each category occurs.
- **Relative frequency table** summarizes the **proportion (fraction)** or **percentage** of occurrences, instead of the count.

Example 2.1 | Recruitment study

- A study was conducted to determine paths to authority for individuals occupying top positions of responsibility in key public-interest organizations. Each of 1,345 individuals was asked for their previous jobs and is classified as one of the four categories: corporate, public-interest, government, other), as appearing in the following dataset.

Employee ID	Recruitment from
100231	Corporate
110978	Public-interest
112003	Government
⋮	⋮
121075	Corporate
123001	Other

- We need to turn “raw” data into [information](#).

Example 2.1 | Summary using Frequency Table

- A **frequency table** shows the **number of cases** that fall in each category.
- The following frequency table shows the number of employees who fall in each category of previous employments.

Recruitment from	Count
Corporate	501
Public-interest	683
Government	94
Other	67
Total	1345

Proportion

- The **proportion** in a category is calculated by

$$\text{proportion} = \frac{\text{number in category}}{\text{total number}}$$

- 1 Proportion for a **population**: p
- 2 Proportion for a **sample**: $\hat{p}$ ("p-hat")

Example 2.1 | Proportion

- What proportion of the employees were recruited from government positions?

Recruitment from	Count
Corporate	501
Public-interest	683
Government	94
Other	67
Total	1345

- We compute

$$\hat{p} = \frac{94}{1345} \approx 0.07 \text{ or } 7\%$$

- Proportions and percentages can be used interchangeably.
- Interpretation: Approximately 7% of the employees were recruited from government positions.

Example 2.1 | Summary using Relative Frequency Table

- A **relative frequency table** shows the **proportion of cases** that fall in each category.
- The following relative frequency table shows the proportion of employees who fall in each category of previous employments.

Recruitment from	Relative frequency
Corporate	0.372
Public-interest	0.508
Government	0.070
Other	0.050
Total	1.000

- All the proportions in a relative frequency table **sum to 1**.

Bar Graph/Chart/Plot

- In a **bar graph**, the height of the bar corresponds to the **number of cases** falling in each category.
- Constructing a bar graph:
 - 1 Label **frequencies** (or **relative frequencies**) on one axis and **categories** of the variable on the other axis.
 - 2 Construct a rectangle at each category of the variable with a height equal to the frequency (or relative frequency) in the category.
 - 3 **Leave a space** between each category to represent distinct and separate categories.

Example 2.1 | Summary using Bar Graph

- Draw a bar graph of the recruitment data.

Recruitment from	Count	Relative freq
Corporate	501	0.372
Public-interest	683	0.508
Government	94	0.070
Other	67	0.050
Total	1345	1.000

- Try **StatKey** at <https://www.lock5stat.com/StatKey/>.

Pie Chart

- In a **pie chart**, the **relative area** of each slice of the pie corresponds to the proportion in each category
- Tips for a “good” pie chart:
 - 1 Choose a small number of categories. (Unnecessarily many categories make the pie chart hard to interpret.)
 - 2 Construct pie chart so that percentages are in either **ascending** or **descending** order.

Example 2.1 | Summary using Pie Chart

- Draw a pie chart of the recruitment data.

Recruitment from	Count	Relative freq
Corporate	501	0.372
Public-interest	683	0.508
Government	94	0.070
Other	67	0.050
Total	1345	1.000

- Try **StatKey** at <https://www.lock5stat.com/StatKey/>.

Two Categorical Variables

- So far, we have looked at a single categorical variable.
- Often, we want to look at **two** categorical variables **together** and see how their outcomes are **related** or **associated**.
 - ➊ To study the relationship between two variables, we observe **both** variables on the **same** individuals.
 - ➋ For example, if we observe the gender and information sources of a large group of people, we know which gender and information sources are related.

Example 2.2 | Gender and Information Source (created)

- A survey was conducted to find the relationship between preferences for television, radio, newspaper, or internet for national news, and gender. The data set from a random sample of 20 students is shown below:

Name	Gender	Source	Name	Gender	Source
Amy	F	Internet	Kathy	F	Newspaper
Arnold	M	TV	Mary	F	Internet
Bill	M	Newspaper	Matthew	M	Internet
David	M	TV	Melissa	F	TV
Edward	M	Radio	Nicole	F	Newspaper
Elizabeth	F	Newspaper	Rebecca	F	Radio
Eric	M	TV	Risa	F	Newspaper
Gretta	F	Internet	Robert	M	TV
James	M	Internet	Viola	F	Internet
John	M	Internet	Zach	M	Internet

- We need to turn “raw” data into information.

Example 2.2 | Summary based on one variable

- A frequency and relative percentage table for the **information source** is

Source	Count	Percent
TV	5	25%
Radio	2	10%
Newspaper	5	25%
Internet	8	40%
total	20	100%

- What do you think?

Two-Way Table

- We look at the **relationship** between two categorical (qualitative) variables.
- When one categorical variable has m categories and the other n categories, a $m \times n$ **two-way table (contingency table)** shows the frequency (or relative frequency) for each **cross** category of the two categorical variables.
- Two-way table can be presented in many different ways.
- It doesn't matter which variable is displayed in the rows and which in the columns.

Example 2.2 | Two-way Frequency Table

- The following 2×4 contingency table shows the *frequency* for each cross category of both of the two variables (gender and information source):

Gender	Information Source				Row Total
	TV	Radio	Newspaper	Internet	
Male	4	1	1	4	10
Female	1	1	4	4	10
Column Total	5	2	5	8	20

- The row and column totals are called **marginal totals** (or **marginals**).
- The total of the marginal totals is the **grand total**.

Example 2.2 | Two-way Relative Frequency Table

- The preceding two-way frequency table may be converted to *percentages* of the grand total by dividing each frequency by the grand total and multiplying by 100:

Gender	Information Source				Row Total
	TV	Radio	Newspaper	Internet	
Male	20%	5%	5%	20%	50%
Female	5%	5%	20%	20%	50%
Column Total	25%	10%	25%	40%	100%

- Interpret these percentages.

Example 2.2 | Two-way Conditional Relative Frequency Table

- The frequencies in the first 2×4 two-way frequency table may also be expressed as *percentages* of the *row totals* (upper) or *column totals* (lower) by dividing each row or column frequency by that row's or column's total and multiplying by 100:

Gender	Information Source				Row Total
	TV	Radio	Newspaper	Internet	
Male	40%	10%	10%	40%	100%
Female	10%	10%	40%	40%	100%
Column Total	25%	10%	25%	40%	100%

Gender	Information Source				Row Total
	TV	Radio	Newspaper	Internet	
Male	80%	50%	20%	50%	50%
Female	20%	50%	80%	50%	50%
Column Total	100%	100%	100%	100%	100%

Example 2.2 | Two-way Table Analysis

■ Questions:

- 1 What proportion of students are female and prefer TV as their primary source of information?
- 2 What proportion of students prefer TV as their primary source of information?
- 3 What proportion of male students prefer TV as their primary source of information?
- 4 What proportion of female students prefer TV as their primary source of information?
- 5 What proportion of students who prefer TV as their primary source of information are male?
- 6 What proportion of students who prefer TV as their primary source of information are female?

Example 2.2 | Two-way Table Analysis (Caution!)

■ CAUTION:

- 1 The proportion of students who are male and prefer TV is **not the same as** the proportion of male students who prefer TV. $20\% \neq 40\%$!
- 2 The proportion of male students who prefer TV is **not the same as** the proportion of TV-preferring students who are male. $40\% \neq 80\%$!
- 3 The proportion of female students who prefer TV is **not the same as** the proportion of TV-preferring students who are female. $10\% \neq 20\%$!

Difference in Proportions

- A **difference in proportions** is a difference in proportions for one categorical variable calculated for different levels of the other categorical variable.
- Example: In the gender and information source example,

Difference in proportions of students preferring TV between males and females

= proportion of males who prefer TV – proportion of females who prefer TV

= $\hat{p}_M - \hat{p}_F$

= 0.40 – 0.10

= 0.30 or 30%

Exercise 2.1 | Two-way table analysis

- A survey was conducted to investigate the relationship between viewers' preferences for ABC, CBS, NBC, or PBS for news information and their political party affiliation.

	ABC	CBS	NBC	PBS
Democrat	200	200	250	150
Republican	450	350	500	200
Other	150	400	100	50

- How many viewers were surveyed?
- Why are these bivariate data? What type of variable is each one?
- How many preferred to watch CBS?
- What percentage of the survey was Republican?
- How many were Democrats and preferred to watch ABC?
- What percentage of the Democrats preferred ABC?
- What is the difference in ABC preference percentages between Democrats and Republicans?

Side-by-Side Bar Chart

- A **side-by-side bar chart** is a type of bar graph used to compare two or more groups across the same categories. Each category has a cluster of bars placed next to each other, with each bar representing a different group.
- Example. Draw a side-by-side bar chart for the gender and information source data.

Segmented Bar Chart

- A **segmented bar chart** is like a side-by-side bar chart, but the bars are stacked instead of side-by-side.
- Example. Draw a segmented bar chart for the gender and information source data.